From State Forecasts to Agent Failure Prediction

Testing the Incremental Value of Task-Relevant Evidence

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Abstract. I investigated whether task-relevant summaries of a model's explicit state forecasts improve prediction of later agent failure beyond replay, task feasibility and likelihood diagnostics. I implemented a frozen public SPA evaluation in native Sokoban, preserving 224 episodes and comparing nine fixed feature views on a family-grouped split. In the final at-risk set (124 episodes; four successes), paired gain intervals crossed zero in both likelihood scopes. Forecasts were available in 25/224 initial packets and exact in 1/224. The study establishes a bounded empirical test and a measurement limitation, rather than a new algorithm or a resolved predictive improvement.

When does a detectable error matter for a decision?

An agent may describe a state plausibly yet fail under its remaining action budget. The aim is to warn about consequential errors, rather than flag every detectable discrepancy. State accuracy, response dependence and action consistency test different properties of a reply. Task relevance concerns goal consequences; failure prediction concerns later outcomes; causal faithfulness concerns whether stated reasoning drives behavior.

Research question. Do task-relevant summaries of explicit state forecasts improve prediction of later agent failure beyond replay, task feasibility and likelihood diagnostics?

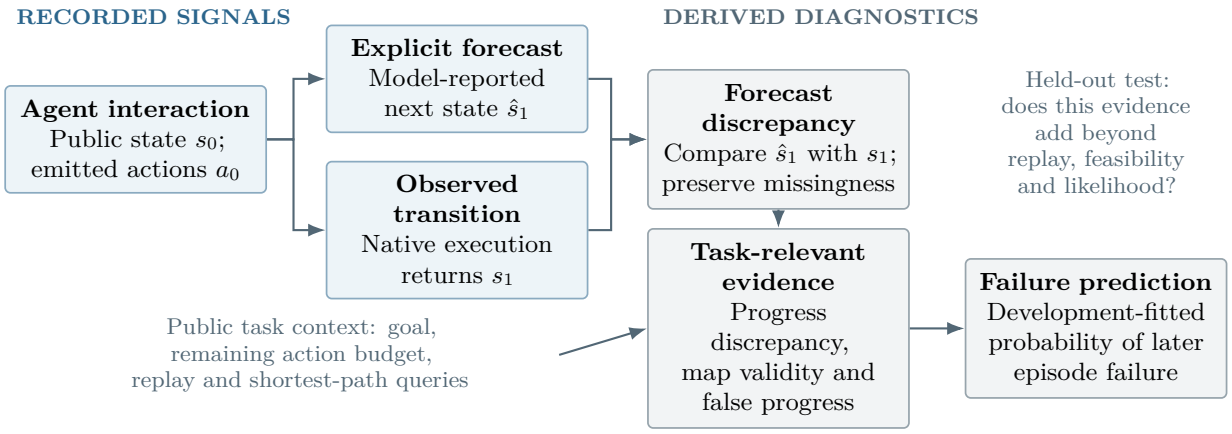


Figure 1. Research question and evidence construction. Blue boxes are recorded signals; slate boxes are derived diagnostic features or fitted predictions. A forecast and an observed transition are compared, then interpreted with the same public goal and budget. Arrows describe evidence flow, not a causal claim about internal reasoning. Missing forecasts are never filled from simulator truth.

What I contributed to this investigation

- A controlled frozen-release experiment with saved prompts, raw tokens, native transitions, fixed feature views and strong replay/task comparisons.
- A paired held-out test that separates a tiny point estimate from a resolved gain, while exposing sparse forecasts, low event counts and an unavailable primary calibration fit.
- A concrete follow-up question: which forecast errors become informative after task feasibility and ordinary replay are already known?

Relation to Professor Li's research

RAGEN-2 already studies reasoning collapse and cross-input diagnostics [1]; VAGEN and SPA study state/transition reasoning or supervision [2,3]. My question concerns a narrower reliability boundary in multi-turn world-model agents: whether explicit forecast errors carry additional *decision-relevant* evidence. This connects to agentic RL and embodied-agent reliability without treating this symbolic task as an embodied-system reproduction.

One controlled pipeline, nine feature views

Execution. Public SPA Sokoban release (1.543B unique parameters), fixed weights; native one-box 6×6 grids. Ten action attempts, five model turns; temperature 0.7, top-p 0.8, top-k 20 and 400 new tokens per response.

Prediction landmark. After the first proposed action batch and returned native observation; zero mapped actions are possible. Failure is no success within the horizon. Exclude cases already complete, terminal or out of budget.

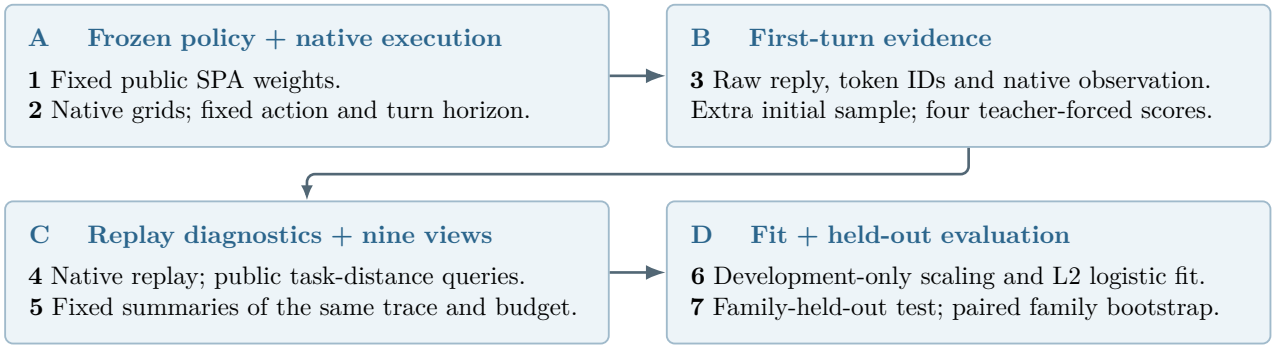


Figure 2. Implemented architecture. Four stages preserve all seven operations. Every view is charged the same recorded evidence packet; this compares summaries at equal spend, not optimal online allocation. All stages were executed.

A split designed around task families

	Development	Calibration	Final
Fixed boards	64	32	128
D4 wall-plus-target families	50	27	106
At-risk episodes	63	30	124
At-risk successes / failures	2 / 61	0 / 30	4 / 120

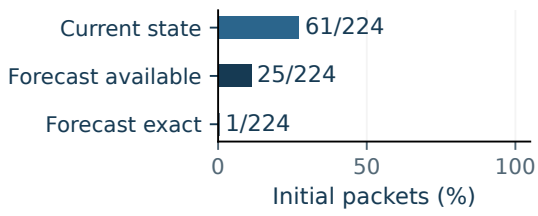
No family crosses splits. Standardization and L2 logistic regression ($C = 1$) use development labels only; no final-set tuning. Family separation does not prove unseen-wall or pretraining-disjoint generalization.

Nine fixed feature views

View	Columns	Evidence used / comparison role
Development prior	0	Smoothed development failure rate.
State + format	11	Current-state scores and missingness.
Ordinary replay	12	Forecast/native agreement; no-op indicator.
Likelihood proxy	9	Own NLL; finite cross-input dependence.
State + replay + likelihood	18	Combined generic diagnostics.
Two-sample disagreement	23	Extra initial-sample disagreement.
Replay + task / deadlock	25	Goal distance, progress and feasibility.
Task-relevance candidate	34	Nine added forecast/task summaries.
Matched generic interactions	34	Nine generic terms as a capacity control.

Nonprior views include format, action, budget and output-length controls. They reuse the same episodes, not nine independent experiments. Equal columns do not guarantee equal effective capacity.

Availability and likelihood are separate measurements



All 224 initial packets. All 25 forecasts are explicit grids; one is exact. Current-state evidence also permits explicit entity coordinates.

Registered: actual generated token IDs through the first closing think tag; this may measure formatting.
Full tail: entire actual first-response tail; same own/three-reference scheme. 896 teacher-forcing passes, zero extra sampling. Planned after collection began, before analysis; added scoring cost is charged. Finite length-normalized proxies, not exact MI or full RAGEN-2 reproductions. Dependence and correctness do not establish causal faithfulness [4].

No resolved increment over the strong baseline

Read with the low-event warning. The final at-risk set has **124 episodes: four successes and 120 failures across 102 layout families**. The primary calibration map cannot be fitted: all 30 at-risk calibration episodes failed. The primary probabilities remain uncalibrated.

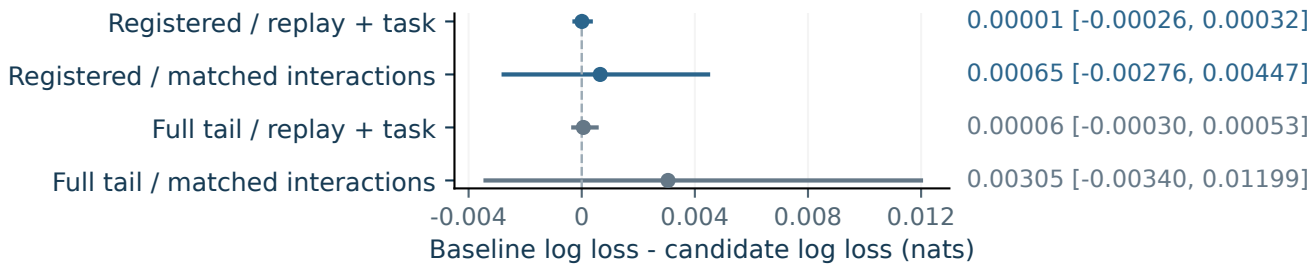


Figure 3. Paired held-out log-loss gain. Positive values favor the task-relevance candidate. Bars are 95% percentile intervals from 2,000 paired wall-plus-target-family resamples, conditional on the development fit. They exclude full retraining uncertainty; comparisons are exploratory. Both replay/task intervals and both matched-interaction intervals cross zero.

View	Held-out log loss ↓		Brier score ↓	
	Registered	Full tail	Registered	Full tail
Development prior	0.1449	0.1449	0.0314	0.0314
State + format	0.1064	0.1064	0.0289	0.0289
Ordinary replay	0.0947	0.0947	0.0272	0.0272
Likelihood proxy	0.1152	0.1028	0.0296	0.0283
State + replay + likelihood	0.1054	0.0920	0.0282	0.0266
Two-sample disagreement	0.1040	0.0901	0.0282	0.0268
Replay + task / deadlock	0.0395	0.0429	0.0097	0.0113
Task-relevance candidate	0.0395	0.0429	0.0096	0.0112
Matched generic interactions	0.0402	0.0459	0.0103	0.0131

What this finding means

The replay/task baseline already captures much of the predictive signal in this cohort. The candidate's additional summaries vary, but their tiny point gains do not resolve an increment. Sparse forecasts (25/224) and rare exact forecasts (1/224) limit the opportunity to measure task-relevant forecast error. This is an inconclusive incremental comparison, not a failed execution, proof of equivalence or evidence that better forecasts cannot help. Novelty remains unestablished (B).

Limits. One small frozen model and deterministic, fully observed Sokoban restrict generalization. Only four final at-risk successes trigger the preset low-event warning. All-episode calibration sensitivity has two successes in 32 calibration cases and does not replace the unavailable primary fit. The inference interface was adapted during qualification; historical generator identity and the RAGEN leaf commit remain unverified. No causal intervention on internal reasoning was performed.

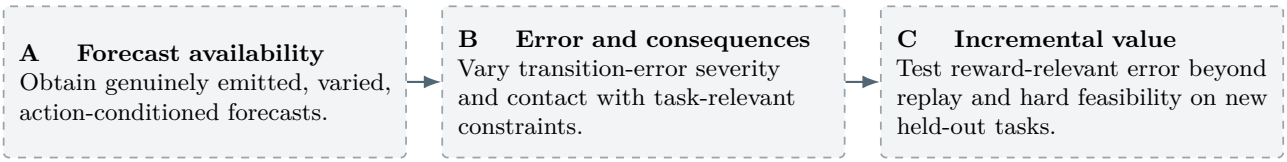


Figure 4. Proposed future research, not completed experiments. First qualify forecast availability and event counts; then test task consequence on a new held-out population with replay/feasibility and matched-information controls. None of these stages has been launched.

Discussion question. Under what conditions does an agent's forecast error become decision-relevant, rather than merely detectable?

References. [1] Z. Wang et al. *RAGEN-2: Reasoning Collapse in Agentic RL.* ICML 2026, PMLR 306. [2] K. Wang et al. *VAGEN.* NeurIPS 2025. [3] S. Chen et al. *Why Do LLM Agents Fail in Exploring New Environments? A World-Modeling Perspective.* arXiv:2510.15047v2. [4] T. Lanham et al. *Measuring Faithfulness in Chain-of-Thought Reasoning.* arXiv:2307.13702v1.

Reproducibility. Editable diagrams, plot code, references, sanitized claim checks and file hashes accompany this note. Token/prompt, native/public replay and independent statistical checks passed. Original 128 records are byte-identical; the resumed 96 retain batches 16–27. No new experiment was run for this document. Public aggregates reproduce figures and tables; raw trajectories and weights are excluded. **Code and public evidence: github.com/ZhijingWu/agent-forecast-failure-prediction**.